Protocol content, ICH M11 and USDM v4

# ICH M11 elements: a target to map to

One global identifier per protocol element, linked to its place in the Template, its specification, its term and its USDM attributes

Data | Data in a string | Words

Kerstin Forsberg · the published ICH M11 Technical Specification, Template and Terminology, and the HL7 and CDISC mappings to USDM v4, represented as a knowledge graph

Two standards, one protocol

# ICH M11 is the document, USDM is the data




**M11 → USDM, as two published mappings state it**

HL7 FHIR Clinical Study Protocol IG ballot, May 2026 — stated against M11 Step 4 and USDM "v4.0.0 (update pending)"

CDISC DDF-RA v4.0.0, m11_mapping.xlsx — stated against the M11 Step 2 draft and USDM v4.0.0

Protocol content needs a place in both. The question is what exactly one maps to: a Template field, a specification block, a code, or a USDM attribute.

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Each mapping is a spreadsheet: a row names an element by code or by name and a USDM class and attribute by name. Data in a string, until it is resolved.

ICH M11 as published

# Not yet a target to map to

Data · Data in a string · Words

## One element, several names

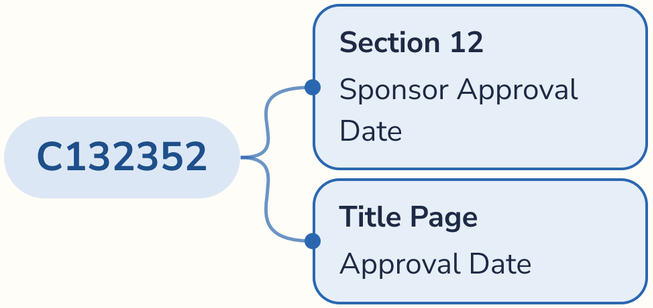
The secondary objective:

| | |
|---|---|
| {<Secondary Objective>} | Specification |
| <Enter Secondary Objective> | Template |
| Secondary Objective | Terminology |

“Back-up Method for Reporting” in the Specification and Terminology is “Backup Method for Reporting” in the Template and in both mappings. Matched by name, that element gets neither its Template field nor a mapping.

Mapping rows with a code that identifies the element: HL7 134 of 324, CDISC 93 of 323. The rest go by name, or not at all.

## A code names a concept, not an element




The element is the pair as the Specification states it: section code list and concept code.

9 codes are elements in two sections: 18 elements.

## The place is written as text

“3.2.X where X is a unique number for each Secondary Objective”

Specification, place in the table of contents

- 3.2.1 the Template, as printed
- 3.2.# the mapping rows

The sections themselves have codes: 160 terms in code list C217272.

Sources: ICH M11 Technical Specification and Template (Step 4, 2025-11-19), ICH M11 Terminology (NCI EVS, 2025-12-19), the HL7 and CDISC mappings; counts from the m11-rdf v0.4.0 build reports.

m11-rdf

# One element, one global identifier, all its links

Data · Data in a string · Words

## Template · 3.2.1 Secondary Objective ↗

<Enter Secondary Objective>

page 29 · text insertion · in { }: conditional

Field linked to the element by m11-rdf, by the Specification term

## Specification · block 193

{<Secondary Objective>}

Text · D · Required · in section 3.2

place: “3.2.X where X is a unique number …”

**Secondary Objective**
C217344.C85827

## Terminology · code C85827 ↗

“The secondary reason for performing a study in terms of the scientific questions to be answered by the analysis of data collected during the study.”

code list C217344, Section 3

ICH OID …2.3.3.3

## USDM v4 · Objective.text ↗

HL7 row 148 · CDISC row 152

where Objective.level = C85827

Both rows, resolved in usdm-rdf: `SyntaxTemplate-text`, the attribute Objective inherits

All links on this slide are in m11-rdf v0.4.0. The Template does not state which element a field is; m11-rdf links the field by the Specification term, and each mapping row by its code or name.

Recorded, not repaired

# Where the standards don't line up, on the same example

## The secondary objective heading

Mapping row, the same in both files (HL7 147, CDISC 151):

Secondary Objective <#> | 3.2.# | Objective.label

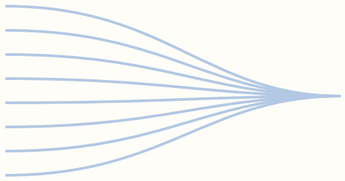
**code C85826 · Primary Objective**

where Objective.level = C85827

**The row contradicts itself:** its code is the primary objective, its condition the secondary.

The Specification has a heading here, not an element: `{3.2.X Secondary Objective <#>}`, H, “Concept: Heading”. The exploratory objective heading has the same pattern (HL7 149, CDISC 153).

## Narrative elements: one attribute, told apart by a section number



**NarrativeContentItem.text**

sectionNumber = <desired section>

277 mapping statements, for 140 elements, point at the same USDM attribute. Only the condition tells them apart, and a section number is a string in the study file.

m11-rdf gives each of the 160 sections a global identifier; 111 Template headings are themselves an element.

And for the field on slide 4: in { } in the Template, Required in the Specification. 4 fields linked to an element are like this.

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m11-rdf keeps each statement as published, flags it, and lists the rest in its build reports (`d2-residue.csv`, `d5-residue.csv`). It does not repair them, and it does not judge which source is right.

What it gives

# Map once, to an M11 element

Map your content to `https://w3id.org/m11-rdf/C217344.C85827` and you get

| | |
|---|---|
| **its place** | the Template section, heading and field, with page |
| **its specification** | term, data type, conformance, cardinality, place in the table of contents |
| **its term** | the NCI EVS code and definition, the ICH OID of its code list |
| **its USDM attributes** | as the HL7 and the CDISC mapping state them, side by side, as usdm-rdf global identifiers |

**270**
elements, from 284 specification blocks, in 16 section code lists

**160**
sections; Template headings on their sections

**281**
Template fields; 271 linked to an element

**496**
mapping statements; 202 elements mapped by both files, all to the same USDM targets

**Not given** a mapping for 55 of the 270 elements; study instances; anything normative. Not an ICH, CDISC or HL7 artifact.

Published sources → m11-rdf (identifiers and a separate mapping set, rebuilt from pinned files) → USDM attributes in usdm-rdf · source: github.com/kerfors/m11-rdf

What is left

# As good as the published sources allow, for now

SETTLED IN M11-RDF

## Identity and place

270 elements, each the pair the Specification states; 160 sections; Template headings on their sections, 111 of them an element.

271 of 281 Template fields linked to an element; 264 of 270 elements reached by a field or a heading.

496 of 647 mapping rows are mapping statements.

NEEDS THE PUBLISHERS

## 151 mapping rows without a statement

98 with a name that is not a Specification term and no usable code. Rows still marked CNEW: HL7 163, CDISC 212.

34 where the name or code fits more than one element, or code and name disagree.

19 where the element is found but the stated USDM class and attribute are not.

Each listed in `d2-residue.csv`; Template residue in `d5-residue.csv`.

NEEDS A CONVENTION

## Content that points at its element

277 statements, for 140 elements, reach USDM only as `NarrativeContentItem.text` plus a section-number string.

A study file needs a way to say: this content is that M11 section or element. Neither USDM v4 nor the mappings have one yet.

m11-rdf applies mechanical, reproducible rules only. Where a rule would need judgement (a near-match of names, a choice between subclasses, a repaired spelling), the row stays in the residue.

USDM v4.1, public review from end of November 2026

# Questions to ask of the alignment with ICH M11

| | |
|---|---|
| **Coverage** | 55 of the 270 elements have no mapping statement. Do they get a place in USDM? |
| **Document content** | Can content point at an M11 section or element, rather than at a section number written as text? |
| **Model and mapping** | 160 rows in each mapping name `StudyProtocolDocumentVersion`, which USDM v4.0.0 does not have (it has StudyDefinitionDocumentVersion). Is the model changed, or the mapping? |
| **Step 4** | The CDISC mapping is stated against the M11 Step 2 draft. For 39 of the 284 section and element-name pairs that both files have, the NCIt code differs. |
| **Codes and names** | Where m11-rdf finds a CNEW row by its name (HL7 128, CDISC 154), the element's code is published in the ICH M11 Terminology. Codes in the cells, and names as in the Specification, would make the mapping resolve by code. |

When 4.1 is published, m11-rdf is rebuilt against it: the same element identifiers, new USDM and mapping inputs, and the difference per element.